Mahika Nair

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EDUCATION

MSc Data Science | University of Edinburgh, United Kingdom *Sep 2025 - Aug 2026*

- Completed August 2026 (On track for Distinction, awaiting final award)
- Current Average: 86.6%

PG Diploma in AI-ML and Leadership | Plaksha University, India *Aug 2023 - Jul 2024*

- CGPA: 9.86/10.0

BSc (Hons.) Mathematics | Krea University, India *Aug 2020 - Jul 2023*

- Minor in Chemistry
- CGPA: 9.24/10.0

WORK EXPERIENCE

Predli Consulting AB | Associate AI Engineer *Sep 2024 - Aug 2025*

- Built an AI-driven document processing and RAG pipeline (LLMs, OCR, vector retrieval) that cut manual processing time from days to minutes (95% reduction).
- Designed agentic AI workflows (AutoGen) for the ResolveWithAI platform, improving response accuracy from 80% to 90% and growing adoption to 10+ organic users within 3 months.
- Built full-stack features (TypeScript) and LangGraph-based agent orchestration for Predli Studio, integrating MCP-compatible tools for autonomous multi-step workflows.

Max Super Specialty Hospital | Summer Research Intern *Jul 2023 - Sep 2023*

- Developed MRI-based tumour segmentation workflows and extracted quantitative radiomics features using PyRadiomics for breast cancer analysis.
- Evaluated radiomic feature importance to identify imaging characteristics associated with tumour heterogeneity and clinical outcomes.

Fractal Analytics | Imagineer Intern *Jun 2022 - Aug 2022*

- Mined and cleaned client sales data to build revenue growth models, supporting data-driven decision-making through structured analysis of large datasets.

Krea University | Summer Research Intern *Jun 2021 - Aug 2021*

- Conducted computational analysis of Ubiquitin and SUMO protein families using structural bioinformatics and molecular visualization tools (Chimera) to investigate evolutionary and structural relationships and their influence on protein function.

PROJECTS

ML Surrogate Modelling for Protein Adsorption | Master's Dissertation *May 2026 - Aug 2026*

- Developed the first deep learning surrogate for Self-Consistent Field Theory (SCFT) protein adsorption simulations, predicting residue-level equilibrium distances with 0.4 nm MAE and achieving a 200× speedup.

- Built a hybrid Transformer and U-Net architecture in PyTorch, integrating coarse-grained sequences, ESM protein language model embeddings, and SCFT targets to learn sequence-dependent adsorption behaviour.
- Established model robustness through ablation, out-of-distribution, and interaction-strength analyses, using GPU-accelerated HPC workflows with **Linux and Slurm**.

Satellite Crop Classification | Capstone Thesis, Plaksha University *Apr 2024 - Jun 2024*

- Collaborated with Professor Shashank Tamaskar to classify crop fields in Punjab using high-resolution Sentinel-2 satellite imagery and Google Earth Engine, providing government stakeholders with timely insights for resource allocation.
- Pretrained a pixel-based vision transformer (PRESTO) on high-resolution Indian satellite data, achieving an F1-score of 85.5% for accurate crop classification.

'The Office' RAG Chatbot | Personal Project *Feb 2024 - Mar 2024*

- Developed a RAG-based chatbot using Streamlit, LangChain, Chroma DB, Gemini, and YouTube API, combining Wikipedia embeddings with relevant video links to answer fan queries with fast semantic retrieval.

Mudra Visualisation with CV | Personal Project *Jan 2024 - Feb 2024*

- Built a YOLOv8 model detecting 30 Indian classical hand gestures in real time (92% accuracy), trained on a self-created dataset of 200 images per class sourced from 35+ individuals.

PUBLICATIONS

- Piyush Sharma, Sneha Thomas, Mahika Nair, and Ananth Govind Rajan, "Machine Learnable and Rapidly Interpretable Language for Nanopores Enables Structure-Property Relationships in Nanoporous 2D Materials", ***Journal of the American Chemical Society*** (2024) [DOI]

SKILLS

- **Programming Languages:** Python (PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV), R, SQL, TypeScript
- **ML/AI:** Transformers, Hugging Face, LLMs, NLP, Computer Vision, RAG, LangChain, LangGraph, AutoGen, MCP, Multi-Agent Systems
- **Infrastructure:** AWS (EC2, ECS, SageMaker), Docker, Linux, Slurm, Git, REST APIs
- **Languages:** English (C2), Hindi (C2)

AWARDS

Plaksha University *2023 - 2024*

- Academic Excellence Award for securing a **top 2 CGPA** in the cohort
- Selected as the Graduation Speaker for the cohort
- 1st place in the startup pitching competition of the Entrepreneurial Challenge Lab

Krea University *2020 - 2023*

- 1st Year Dean's List student